

Last login: Mon Jun 15 14:59:00 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# -- Step 1: Reset role and disable RLS
RESET ROLE;
ALTER TABLE orders DISABLE ROW LEVEL SECURITY;

-- Step 2: Drop existing view if any
DROP VIEW IF EXISTS orders_secure;

-- Step 3: Create view with same filter logic as RLS policy
CREATE VIEW orders_secure AS

```
SELECT o.*
FROM orders o
WHERE EXISTS (
  SELECT 1 FROM customer c
  WHERE c.c_custkey = o.o_custkey
  AND c.c_acctbal > 5000
);
```

-- Step 4: Grant permissions and set role
GRANT SELECT ON orders_secure TO alice;
GRANT SELECT ON customer TO alice;
GRANT SELECT ON lineitem TO alice;
SET ROLE alice;
RESET
ALTER TABLE
DROP VIEW
CREATE VIEW
GRANT
GRANT
GRANT
SET

tpch=> EXPLAIN (
 ANALYZE true,
 VERBOSE true,
 COSTS true,
 BUFFERS false,
 TIMING false,
 FORMAT JSON
)

```
select
  o_orderpriority,
  count(*) as order_count
from
  orders_secure
where
  o_orderdate >= date '1994-01-01'
  and o_orderdate < date '1994-01-01' + interval '3' month
  and exists (
    select
      *
    from
      lineitem
    where
      l_orderkey = o_orderkey
      and l_commitdate < l_receiptdate
  )
group by
  o_orderpriority
order by
  o_orderpriority;
```

WARNING: ===== planner input query tree =====

```
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTY
:alias
  {ALIAS
:aliasname o
:colnames <>
}
}
:eref
  {ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
```

```

)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opertName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityTmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtkind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST

```

c.c_custkey = o.o_custkey

c.c_acctbal > 5000

```

        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ -120 19 0 0 0 0 0 0 ]
    }
    }
    :location -1
    }
    }
    :location -1
    }
    }
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname o_orderkey
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 1
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
    :resno 2
    :resname o_custkey
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 2
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 3
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    :resno 3
    :resname o_orderstatus
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 3
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 4
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16950
:inh false

```

```

:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:erefs
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 116
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16950
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtrindex 1
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 194
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 214
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 206
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 235
}
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 254
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 278
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 267
}
)
:location 247
}
{SUBLINK
:subLinkType 0

```

o.o_orderdate >= DATE '1994-01-01'

o.o_orderdate < DATE '1994-01-01'
+ INTERVAL '3 month'

```

:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 394
}
)
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 407
}
)
:location 405
}
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 434
}
)
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 449
}
)
:location 447
}
)
:location 430
}
}
:mergeActionList <>

```

l.l_orderkey = o.o_orderkey

l.l_commitdate <
l.l_receiptdate

```

:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 332
}
:resno 1
:resname 1_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 332
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 332
}
:resno 3
:resname 1_supplekey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 332
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 332
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 332
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7

```

```

        :location 332
    }
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 332
}
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 332
}
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 332
}
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 332
}
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 332
}
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 332
}
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1

```



```

        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 14
        :location 332
    }
    :resno 14
    :resname l_shipinstruct
    :ressortgrouppref 0
    :resorigtbl 16566
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location 332
    }
    :resno 15
    :resname l_shipmode
    :ressortgrouppref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 332
    }
    :resno 16
    :resname l_comment
    :ressortgrouppref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    }
    :location 296
    }
    :location 231
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 1
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgrouppref 1
    :resorigtbl 16950
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 137
    }
    :resno 2
    :resname order_count
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    }
    :override 0
    :onConflict <>

```

```

:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
   :tleSortGroupRef 1
   :eqop 98
   :sortop 664
   :reverse_sort false
   :nulls_first false
   :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
   :tleSortGroupRef 1
   :eqop 98
   :sortop 664
   :reverse_sort false
   :nulls_first false
   :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: pull_up_sublinks - File: src/backend/optimizer/prop/propjointree.c, Relation: N/A, Explanation: Initiating pull-up of sublinks (ANY/EXISTS)

WARNING: pull_up_sublinks - File: src/backend/optimizer/prop/propjointree.c, Relation: N/A, Explanation: Initiating pull-up of sublinks (ANY/EXISTS)

WARNING: ===== after pull up and before planning path list =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
   :alias <>
   :eref
   {ALIAS
    :aliasname orders_secure
    :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
   }
   :rtekind 1
   :subquery <>
   :security_barrier false
   :relid 16950
   :inh false
   :relkind v
   :rellockmode 1
   :perminfoindex 1
   :lateral false
   :inFromCl true
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias <>
   :eref
   {ALIAS
    :aliasname *GROUP*
    :colnames ("o_orderpriority")
   }
   :rtekind 9
   :groupexprs (
    {VAR
     :varno 4
     :varattno 6
     :vartype 25
     :vartypmod -1
     :varcollid 100
     :varnullingrels (b)
     :varlevelsup 0
     :varreturningtype 0
     :varnosyn 4
     :varattnosyn 6
     :location -1
    }
   )
   :lateral false
   :inFromCl false
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias <>
   :eref
   {ALIAS
    :aliasname lineitem
    :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
   }
   :rtekind 0
   :relid 16566
   :inh false
   :relkind r
   :rellockmode 1
   :perminfoindex 2
   :tablesample <>
   :lateral false
   :inFromCl true
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias
   {ALIAS
    :aliasname o
    :colnames <>
   }
   :eref
   {ALIAS
    :aliasname o
    :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
   }
   :rtekind 0
   :relid 16561
   :inh false
   :relkind r
   :rellockmode 1
   :perminfoindex 3
   :tablesample <>
  }
}

```

```

:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesampl <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16950
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals <>
}
}
:rarg
{RANGETBLREF
:rtindex 5
}
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
)
:location -1
}
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
}
)
}
}
)
}
)

```

orders o
SEMI JOIN customer c
ON c.c_custkey =
o.o_custkey
AND c.c_acctbal >
5000

```

:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
}
:alias <>
:rtindex 0
}
)
:quals (
{OEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 214
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 206
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 247
}
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 394
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
:location 405
}
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12

```

o.o_orderdate >= DATE
'1994-01-01'

o.o_orderdate <
'1994-04-01 00:00:00'

lineitem.l_orderkey =
orders.o_orderkey

lineitem.l_commitdate <
lineitem.l_receiptdate

```
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 434
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 449
}
}
:location 447
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16950
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 137
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: === PLANNER SUGGESTION ===
Plan:
Finalize GroupAggregate (cost=194594.52..194617.59 rows=5 width=17)
Output: o.o_orderpriority, count(*)
Group Key: o.o_orderpriority
-> Gather Merge (cost=194594.52..194617.48 rows=12 width=17)
Output: o.o_orderpriority, (PARTIAL count(*))
Workers Planned: 2
-> Partial GroupAggregate (cost=193594.49..193616.07 rows=5 width=17)
Output: o.o_orderpriority, PARTIAL count(*)
Group Key: o.o_orderpriority
-> Sort (cost=193594.49..193601.67 rows=2870 width=9)
Output: o.o_orderpriority
Sort Key: o.o_orderpriority
-> Parallel Hash Semi Join (cost=154455.47..193429.66 rows=2870 width=9)
Output: o.o_orderpriority
```

```
Hash Cond: (o.o_orderkey = lineitem.l_orderkey)
-> Parallel Hash Semi Join (cost=4806.41..40362.28 rows=10210 width=13)
    Output: o.o_orderpriority, o.o_orderkey
    Hash Cond: (o.o_custkey = c.c_custkey)
    -> Parallel Seq Scan on public.orders o (cost=0.00..35383.00 rows=22585 width=17)
        Output: o.o_orderkey, o.o_custkey, o.o_orderstatus, o.o_totalprice, o.o_orderdate, o.o_orderpriority, o.o_clerk, o.o_shippriority, o.o_comment
        Filter: ((o.o_orderdate >= '1994-01-01'::date) AND (o.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
    -> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4)
        Output: c.c_custkey
        -> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4)
            Output: c.c_custkey
            Filter: (c.c_acctbal > '5000'::double precision)
    -> Parallel Hash (cost=135974.79..135974.79 rows=833461 width=4)
        Output: lineitem.l_orderkey
        -> Parallel Seq Scan on public.lineitem (cost=0.00..135974.79 rows=833461 width=4)
            Output: lineitem.l_orderkey
            Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)

=====
WARNING: === EXECUTOR EXECUTION ===
Rows Processed: 5
Plan & Stats:
Finalize GroupAggregate (cost=194594.52..194617.59 rows=5 width=17) (actual time=1487.572..1517.327 rows=5.00 loops=1)
    Output: o.o_orderpriority, count(*)
    Group Key: o.o_orderpriority
    Buffers: shared hit=16 read=134400, temp read=20115 written=20512
    -> Gather Merge (cost=194594.52..194617.48 rows=12 width=17) (actual time=1487.313..1517.300 rows=15.00 loops=1)
        Output: o.o_orderpriority, (PARTIAL count(*))
        Workers Planned: 2
        Workers Launched: 2
        Buffers: shared hit=16 read=134400, temp read=20115 written=20512
        -> Partial GroupAggregate (cost=193594.49..193616.07 rows=5 width=17) (actual time=1478.832..1479.666 rows=5.00 loops=3)
            Output: o.o_orderpriority, PARTIAL count(*)
            Group Key: o.o_orderpriority
            Buffers: shared hit=16 read=134400, temp read=20115 written=20512
            Worker 0: actual time=1476.022..1476.852 rows=5.00 loops=1
                Buffers: shared hit=8 read=44737, temp read=6628 written=6796
            Worker 1: actual time=1475.425..1476.205 rows=5.00 loops=1
                Buffers: shared hit=8 read=44483, temp read=6748 written=6820
            -> Sort (cost=193594.49..193601.67 rows=2870 width=9) (actual time=1478.626..1479.014 rows=7772.33 loops=3)
                Output: o.o_orderpriority
                Sort Key: o.o_orderpriority
                Sort Method: quicksort Memory: 193kB
                Buffers: shared hit=16 read=134400, temp read=20115 written=20512
                Worker 0: actual time=1475.800..1476.186 rows=8012.00 loops=1
                    Sort Method: quicksort Memory: 193kB
                    Buffers: shared hit=8 read=44737, temp read=6628 written=6796
                Worker 1: actual time=1475.229..1475.596 rows=7458.00 loops=1
                    Sort Method: quicksort Memory: 193kB
                    Buffers: shared hit=8 read=44483, temp read=6748 written=6820
                -> Parallel Hash Semi Join (cost=154455.47..193429.66 rows=2870 width=9) (actual time=1326.361..1475.982 rows=7772.33 loops=3)
                    Output: o.o_orderpriority
                    Hash Cond: (o.o_orderkey = lineitem.l_orderkey)
                    Buffers: shared read=134400, temp read=20115 written=20512
                    Worker 0: actual time=1312.984..1472.740 rows=8012.00 loops=1
                        Buffers: shared read=44737, temp read=6628 written=6796
                    Worker 1: actual time=1327.913..1472.769 rows=7458.00 loops=1
                        Buffers: shared read=44483, temp read=6748 written=6820
                    -> Parallel Hash Semi Join (cost=4806.41..40362.28 rows=10210 width=13) (actual time=76.045..231.450 rows=8461.00 loops=3)
                        Output: o.o_orderpriority, o.o_orderkey
                        Hash Cond: (o.o_custkey = c.c_custkey)
                        Buffers: shared read=29680
                        Worker 0: actual time=76.216..231.859 rows=8546.00 loops=1
                            Buffers: shared read=9983
                        Worker 1: actual time=76.184..231.213 rows=8399.00 loops=1
                            Buffers: shared read=9768
                        -> Parallel Seq Scan on public.orders o (cost=0.00..35383.00 rows=22585 width=17) (actual time=0.743..148.301 rows=18783.00 loops=3)
                            Output: o.o_orderkey, o.o_custkey, o.o_orderstatus, o.o_totalprice, o.o_orderdate, o.o_orderpriority, o.o_clerk, o.o_shippriority, o.o_comment
                            Filter: ((o.o_orderdate >= '1994-01-01'::date) AND (o.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
                            Rows Removed by Filter: 481217
                            Buffers: shared read=26008
                            Worker 0: actual time=1.019..148.583 rows=19012.00 loops=1
                                Buffers: shared read=8734
                            Worker 1: actual time=0.679..148.143 rows=18756.00 loops=1
                                Buffers: shared read=8606
                        -> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4) (actual time=75.058..75.058 rows=22663.00 loops=3)
                            Output: c.c_custkey
                            Buckets: 131072 Batches: 1 Memory Usage: 3744kB
                            Buffers: shared read=3672
                            Worker 0: actual time=74.862..74.862 rows=23278.00 loops=1
                                Buffers: shared read=1249
                            Worker 1: actual time=75.155..75.155 rows=21524.00 loops=1
                                Buffers: shared read=1162
                            -> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4) (actual time=0.957..70.273 rows=22663.00 loops=3)
                                Output: c.c_custkey
                                Filter: (c.c_acctbal > '5000'::double precision)
                                Rows Removed by Filter: 27337
                                Buffers: shared read=3672
                                Worker 0: actual time=0.854..70.153 rows=23278.00 loops=1
                                    Buffers: shared read=1249
                                Worker 1: actual time=1.165..70.465 rows=21524.00 loops=1
                                    Buffers: shared read=1162
                        -> Parallel Hash (cost=135974.79..135974.79 rows=833461 width=4) (actual time=1071.737..1071.738 rows=1264432.00 loops=3)
                            Output: lineitem.l_orderkey
                            Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6784kB
                            Buffers: shared read=104720, temp read=9111 written=20128
                            Worker 0: actual time=1068.847..1068.848 rows=1258534.00 loops=1
                                Buffers: shared read=34754, temp read=3001 written=6660
                            Worker 1: actual time=1068.855..1068.856 rows=1257862.00 loops=1
                                Buffers: shared read=34715, temp read=3037 written=6692
                            -> Parallel Seq Scan on public.lineitem (cost=0.00..135974.79 rows=833461 width=4) (actual time=0.667..576.111 rows=1264432.00 loops=3)
                                Output: lineitem.l_orderkey
                                Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
                                Rows Removed by Filter: 735973
                                Buffers: shared read=104720
                                Worker 0: actual time=0.212..574.156 rows=1258534.00 loops=1
                                    Buffers: shared read=34754
                                Worker 1: actual time=0.205..577.636 rows=1257862.00 loops=1
                                    Buffers: shared read=34715
```

```
=====
QUERY PLAN
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    "Plan": {
      "Node Type": "Aggregate",
      "Strategy": "Sorted",
      "Partial Mode": "Finalize",
      "Parallel Aware": false,
      "Async Capable": false,
      "Startup Cost": 194594.52,
      "Total Cost": 194617.59,
      "Plan Rows": 5,
      "Plan Width": 17,
      "Actual Rows": 5.00,
      "Actual Loops": 1,
      "Disabled": false,
      "Output": ["o.o_orderpriority", "count(*)"],
      "Group Key": ["o.o_orderpriority"],
      "Plans": [
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          {
            "Node Type": "Gather Merge",
            "Parent Relationship": "Outer",
            "Parallel Aware": false,
            "Async Capable": false,
            "Startup Cost": 194594.52,
            "Total Cost": 194617.48,
            "Plan Rows": 12,
            "Plan Width": 17,
            "Actual Rows": 15.00,
            "Actual Loops": 1,
            "Disabled": false,
            "Output": ["o.o_orderpriority", "(PARTIAL count(*))"],
            "Workers Planned": 2,
            "Workers Launched": 2,
            "Plans": [
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                  "Partial Mode": "Finalize",
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                  "Async Capable": false,
                  "Startup Cost": 194594.52,
                  "Total Cost": 194617.59,
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                  "Plan Width": 17,
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                  "Group Key": ["o.o_orderpriority"],
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                        "Strategy": "Hash",
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                        "Startup Cost": 193594.49,
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                        "Plan Rows": 5,
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                              "Async Capable": false,
                              "Startup Cost": 193594.49,
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                              "Plan Rows": 2870,
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                              "Buffers": {
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                                  "hit": 16,
                                  "read": 134400,
                                  "temp": {
                                    "read": 20115,
                                    "written": 20512
                                  }
                                },
                                "temp": {
                                  "read": 20115,
                                  "written": 20512
                                }
                              },
                              "Workers": {
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                                "Launched": 1,
                                "Actual": {
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                                    "rows": 5.00,
                                    "loops": 1,
                                    "buffers": {
                                      "shared": {
                                        "hit": 8,
                                        "read": 44737,
                                        "temp": {
                                          "read": 6628,
                                          "written": 6796
                                        }
                                      }
                                    }
                                  },
                                  "1": {
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                                      "shared": {
                                        "hit": 8,
                                        "read": 44483,
                                        "temp": {
                                          "read": 6748,
                                          "written": 6820
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                                      }
                                    }
                                  }
                                }
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                            },
                            {
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                                    "buffers": {
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                                          "written": 6796
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                            },
                            {
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                              "Plan Rows": 10210,
                              "Plan Width": 13,
                              "Actual Rows": 8461.00,
                              "Actual Loops": 3,
                              "Disabled": false,
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                              "Hash Cond": ["o.o_custkey = c.c_custkey"],
                              "Buffers": {
                                "shared": {
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                                "temp": {
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                                  "written": 9768
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                                  "0": {
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                                    }
                                  },
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                                }
                              }
                            },
                            {
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                              "Filter": ["((o.o_orderdate >= '1994-01-01'::date) AND (o.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))"],
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                                    "loops": 1,
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                                        "read": 8734
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                              "Output": ["lineitem.l_orderkey"],
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                                },
                                "temp": {
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                              },
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                                "Launched": 1,
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                                    "buffers": {
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                                        "temp": {
                                          "read": 3037,
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                                "temp": {
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                                  "written": 20128
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                              },
                              "Workers": {
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                                    }
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                                  "1": {
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                                  }
                                }
                              }
                            }
                          ]
                        }
                      ]
                    }
                  ]
                }
              ]
            }
          ]
        }
      ]
    }
  ]
}
```

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"Disabled": false,
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  {
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    "Actual Rows": 5.00,
    "Actual Loops": 1
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"Plans": [
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    "Actual Loops": 3,
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    "Sort Space Used": 193,
    "Sort Space Type": "Memory",
    "Workers": [
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    "Hash Cond": "(o.o_orderkey = lineitem.l_orderkey)",
    "Workers": [
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    ]
  },
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    "Node Type": "Hash Join",
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    "Parallel Aware": true,
    "Async Capable": false,
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    "Startup Cost": 4806.41,
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    "Plan Rows": 10210,
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    "Actual Loops": 3,
    "Disabled": false,
    "Output": ["o.o_orderpriority", "o.o_orderkey"],
    "Inner Unique": false,
    "Hash Cond": "(o.o_custkey = c.c_custkey)",
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      {
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        "Actual Rows": 8399.00,
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    ]
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  {
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    "Async Capable": false,
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    "Schema": "public",
    "Alias": "o",
    "Startup Cost": 0.00,
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    "Plan Rows": 22585,
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    "Actual Loops": 3,
    "Disabled": false,
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    "Filter": "((o.o_orderdate >= '1994-01-01'::date) AND (o.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))",
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    "Workers": [
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        "Actual Loops": 1
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    ]
  }
]
}

```

```

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      "Original Hash Buckets": 131072,
      "Hash Batches": 1,
      "Original Hash Batches": 1,
      "Peak Memory Usage": 3744,
      "Workers": [
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          "Worker Number": 0,
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          "Actual Loops": 1
        }
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      {
        "Worker Number": 1,
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        "Actual Loops": 1
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    },
    "Plans": [
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        "Parent Relationship": "Outer",
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        "Async Capable": false,
        "Relation Name": "customer",
        "Schema": "public",
        "Alias": "c",
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        "Plan Rows": 28253,
        "Plan Width": 4,
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        "Actual Loops": 3,
        "Disabled": false,
        "Output": [{"c.c_custkey"}],
        "Filter": "(c.c_acctbal > '5000'::double precision)",
        "Rows Removed by Filter": 27337,
        "Workers": [
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            "Worker Number": 0,
            "Actual Rows": 23278.00,
            "Actual Loops": 1
          },
          {
            "Worker Number": 1,
            "Actual Rows": 21524.00,
            "Actual Loops": 1
          }
        ]
      }
    ]
  },
  {
    "Node Type": "Hash",
    "Parent Relationship": "Inner",
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    "Startup Cost": 135974.79,
    "Total Cost": 135974.79,
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    "Plan Width": 4,
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    "Actual Loops": 3,
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    "Hash Batches": 32,
    "Original Hash Batches": 16,
    "Peak Memory Usage": 6784,
    "Workers": [
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        "Actual Rows": 1258534.00,
        "Actual Loops": 1
      },
      {
        "Worker Number": 1,
        "Actual Rows": 1257862.00,
        "Actual Loops": 1
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  },
  "Plans": [
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      "Parent Relationship": "Outer",
      "Parallel Aware": true,
      "Async Capable": false,
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      "Schema": "public",
      "Alias": "lineitem",
      "Startup Cost": 0.00,
      "Total Cost": 135974.79,
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      "Plan Width": 4,
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          "Actual Loops": 1
        },
        {
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          "Actual Loops": 1
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      ]
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},
"Planning Time": 16.877,
"Triggers": [
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"Execution Time": 1520.942

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